9.1 Circle Vocabulary & Central Angles

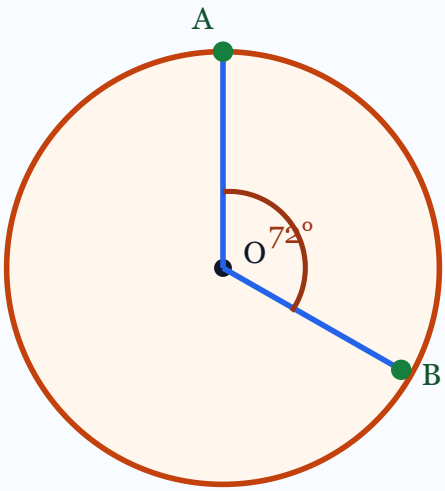
A circle is the set of all points a fixed distance (the **radius**) from a center point. A **chord** connects any two points on the circle; a **diameter** is a chord through the center and is the longest possible chord, with length $2r$. A **central angle** has its vertex at the center, and its measure always equals the measure of the arc it intercepts — this is the single most important rule in circle geometry, since it converts angle problems into arc problems and back. Arcs smaller than a semicircle are **minor arcs**; arcs larger are **major arcs**; an arc cut off by a diameter is a **semicircle** ($180°$). The **Arc Addition Postulate** works just like the Segment and Angle Addition Postulates: adjacent arcs add up. Since central angles all the way around a point sum to $360°$, so do all the arcs of a circle. This worksheet uses these rules together, including algebra and real-world angle problems.

Examples

Example 1: Minor and major arcs

Central angle $\angle AOB = 72°$. Find the measures of minor arc AB and major arc AB.

Solution



The central angle and its intercepted (minor) arc always have equal measure.

Minor arc equals the central angle: $\overset{\frown}{mAB} = 72°$.

Major arc is the rest of the circle: $360° - 72° = 288°$.

Answer: minor arc $= 72°$; major arc $= 288°$.

Example 2: Arc Addition Postulate

$m\overset{\frown}{AB} = 40°$ and $m\overset{\frown}{BC} = 65°$, with B between A and C on the circle. Find $m\overset{\frown}{AC}$.

Solution

Adjacent arcs add, just like adjacent angles:

$m\overset{\frown}{AC} = m\overset{\frown}{AB} + m\overset{\frown}{BC} = 40° + 65° = 105°$.

Answer: $105°$.

Example 3: Central angles around a point sum to 360°

Three central angles around a center measure $110°$, $95°$, and x. Find x.

Solution

Identify the relationship: all central angles around a point sum to $360°$.

Set up and solve the equation:

$110° + 95° + x = 360°$

$205° + x = 360°$

$x = 155°$.

Answer: $155°$.

Example 4: Diameter forms a linear pair of central angles

A and B are endpoints of a diameter; C is another point on the circle. $m\angle AOC = 2x$ and $m\angle BOC = 3x$. Find x and each angle.

Solution

$\angle AOC$ and $\angle BOC$ form a straight (linear) angle along the diameter, so they sum to $180°$:

$2x + 3x = 180°$

$5x = 180° \Rightarrow x = 36°$.

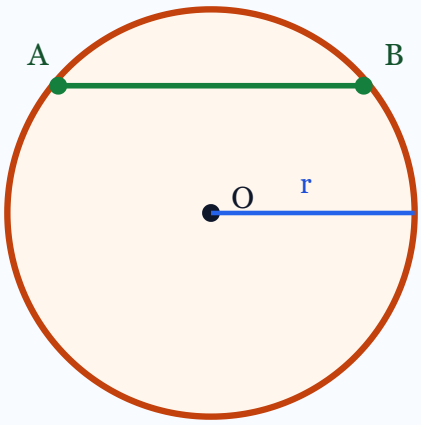
$m\angle AOC = 2(36) = 72°$; $m\angle BOC = 3(36) = 108°$.

Answer: $x = 36°$; angles $72°$ and $108°$.

Example 5: Testing the longest-chord rule

A circle has radius 9. Is a chord of length 20 possible?

Solution



No chord can be longer than the diameter — the diameter is the longest chord.

The diameter is $2(9) = 18$, and no chord can exceed the diameter.
Since $20 > 18$, **no** such chord is possible.

Example 6: Radius from diameter

A circle has diameter 14. Find the radius.

Solution

Identify the given value: $d = 14$.

The radius is always half the diameter:

$$r = \frac{d}{2} = \frac{14}{2} = 7.$$

Answer: 7.

Example 7: Central angle from an arc

Minor arc PQ measures $130°$. Find the central angle that intercepts it.

Solution

A central angle equals its intercepted arc: $m\angle POQ = 130°$.

Answer: $130°$.

Example 8: Semicircle

Chord $\overline{AB}$ is a diameter of the circle. What is $m\widehat{AB}$ (the arc on either side)?

Solution

A diameter divides the circle into two semicircles, each measuring $180°$: $m\widehat{AB} = 180°$.

Answer: $180°$.

Example 9: Chord vs. diameter

A circle has radius 6.5. What is the longest possible chord?

Solution

The diameter is the longest chord: $d = 2(6.5) = 13$.

Answer: 13.

Example 10: Central angles in a given ratio

Four central angles around a point are in the ratio $1 : 2 : 3 : 4$. Find each angle.

Solution

Let the angles be x, $2x$, $3x$, $4x$. They sum to $360°$:

$x + 2x + 3x + 4x = 360°$

$10x = 360° \Rightarrow x = 36°$.

Angles: $36°$, $72°$, $108°$, $144°$.

Check: $36 + 72 + 108 + 144 = 360°$. ✓

Example 11: Application — a clock face

A clock face is a circle with the center at the pivot of its hands. Find the central angle (and the corresponding arc) from the 12 to the 4.

Solution

A clock's 12 hour marks divide the circle into 12 equal central angles: each is $\frac{360^\circ}{12} = 30^\circ$.

From 12 to 4 spans 4 of these equal angles: $4(30^\circ) = 120^\circ$.

Since the central angle equals its intercepted arc, the arc from 12 to 4 also measures 120°.

Answer: 120°.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Minor and major arcs

A central angle measures $130°$. Find the minor and major arc measures.

Answer: **Verification:**

Problem 2: Arc Addition

$\overset{\frown}{mAB} = 52°$, $\overset{\frown}{mBC} = 88°$. Find $\overset{\frown}{mAC}$.

Answer: **Verification:**

Problem 3: Sum to 360°

Three central angles measure $100°$, $120°$, and x. Find x.

Answer: **Verification:**

Problem 4: Diameter linear pair

A, B are diameter endpoints; $m\angle AOC = 4x$ and $m\angle BOC = 5x$. Find x and each angle.

Answer: **Verification:**

Problem 5: Radius and diameter

A circle has radius 6.5. Find its diameter.

Answer: **Verification:**

Problem 6: Longest chord

A circle has radius 8. Can it contain a chord of length 17?

Answer: **Verification:**

Problem 7: Central from arc

Minor arc CD measures $95°$. Find the central angle intercepting it.

Answer: **Verification:**

Problem 8: Major arc

Central angle $\angle XOY = 48°$. Find the major arc XY.

Answer: **Verification:**

Problem 9: Arc Addition

$m\widehat{PQ} = 75°$ and $m\widehat{QR} = 55°$. Find $m\widehat{PR}$.

Answer: **Verification:**

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Problem 10: Semicircle

What is the measure of a semicircle's arc?

Answer: **Verification:**

Problem 11: Sum to 360°

Four equal central angles fill a circle. Find each measure.

Answer: **Verification:**

Problem 12: Diameter

A circle has diameter 22. Find the radius.

Answer: **Verification:**

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Problem 13: Vocabulary

Is a diameter also a chord? Explain briefly.

Answer: **Verification:**

Problem 14: Challenge

Four central angles around a point are in the ratio $2:3:4:6$. Find each angle.

Answer: **Verification:**

Problem 15: Challenge

A pizza is cut into 5 slices whose central angles form an arithmetic sequence with common difference $8°$ (each slice $8°$ bigger than the last). Find the smallest and largest slice angles.

Answer: **Verification:**

✓ Answer Key

1. Minor 130°; major 230°
2. 140°
3. $x = 140°$
4. $x = 20°$; angles 80° and 100°
5. 13
6. No ($d = 16$)
7. 95°
8. 312°
9. 130°
10. 180°
11. 90°
12. 11
13. Yes — it's the longest possible chord, since it also connects two points on the circle
14. 48°, 72°, 96°, 144°
15. Smallest 56°, largest 88°

Great work — keep practising!